

Product Market Dynamics over the Business Cycle

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Abstract

Business cycles reshape product market dynamics, which exhibit substantial heterogeneity across markets. This heterogeneity arises from differences in demand structures and their interaction with entry and exit dynamics. Previous analyses have largely overlooked how product market dynamics are shaped by heterogeneous consumer demand. This paper examines how product entry and exit over the business cycle depend on the structure of consumer demand. I develop a model of strategic producer dynamics under imperfect competition with both heterogeneous producers and consumers to capture these interactions. Using detailed scanner data from the consumer goods sector, I estimate the model for each differentiated product market to analyze producer responses during the Great Recession. The findings reveal that producers respond to negative demand shocks by reducing entry and increasing exit, with the magnitude of these responses varying systematically with the price elasticity of demand. The effects are stronger in markets with less elastic demand, and the interaction between demand elasticity and product characteristics further amplifies the effects of recessions. The results also show that the welfare costs of the business cycle are substantially larger when accounting for demand heterogeneity, with most of the losses stemming from declines in consumer surplus. In the absence of consumer heterogeneity, the recovered welfare loss is significantly understated.

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1 Introduction

How do business cycles reshape product market dynamics through product entry and exit decisions? This paper addresses this question by examining how heterogeneous consumer demand interacts with producers' strategic dynamic decisions to shape welfare over the business cycle. At the aggregate level, fluctuations in entry and exit generate a clear cyclical pattern in product variety (Broda and Weinstein, 2010; Argente, Lee, and Moreira, 2018), but this relationship masks substantial heterogeneity across the product space. For example, the number of products in the canned soup market fell by about 15%, whereas markets such as jams and bottled water experienced only modest declines of around 1.5% following the Great Recession. This heterogeneity indicates that recessions do not uniformly reshape the product space, thereby generating systematic differences in welfare implications. However, relatively little attention has been paid to understanding these micro-level differences.

In this paper, I develop a structural framework that links heterogeneous consumer demand to producers' dynamic entry and exit decisions over the business cycle. This framework allows me to quantify how changes in product market dynamics amplify the welfare costs of the business cycle, costs that are substantially understated when demand heterogeneity is ignored. My approach extends dynamic oligopoly frameworks by adopting a state-of-the-art demand system with heterogeneous consumers. This extension enables the analysis of product market dynamics over the business cycle within a tractable equilibrium framework. By estimating this model, I provide new micro-level evidence on how product entry and exit respond to aggregate demand shocks—an important source of business cycle fluctuations—across markets with distinct demand structures.

Analyzing this interaction between heterogeneous demand structures and producers' entry and exit dynamics poses two main challenges. First, studying detailed product-level dynamics requires the use of rich microdata, which can reveal heterogeneity that is obscured at the aggregate level. However, the granularity of such data also makes it difficult to integrate into a macroeconomic framework. Second, producers' entry and exit decisions are dynamic and strategic: they must track competitors' states and actions when making their own decisions. This interdependence creates a high-dimensional state space and substantial computational complexity.

In this paper, I address these challenges by integrating tools from both macroeconomics and industrial organization. Specifically, I incorporate a canonical discrete-choice demand framework from industrial organization (Berry, Levinsohn, and Pakes, 1995) using framework of Salanié and Wolak (2022) into a model of strategic producer dynamics (Ericson and Pakes, 1995; Ifrach and Weintraub, 2017). This integrated framework offers two key advantages. First, it allows for flexible substitution patterns grounded in microdata through consumer heterogeneity, which cannot be captured in models that assume a representative demand structure (Nevo, 2011). Second, the moment-based approach of Ifrach and Weintraub (2017) preserves the key strategic forces of a Markov Perfect Equilibrium while greatly simplifying computation through the Moment-Based Markov Equilibrium framework, in the spirit of Krusell and Smith (1998) and Hopenhayn (1992). I then estimate this model using highly disaggregated product-level scanner data from the consumer goods sector, recovering demand parameters, a state variable summarizing each product’s demand shifter, and dynamic parameters.¹ I finally conduct counterfactual analyses to quantify the welfare costs of business cycle and to assess the role of demand heterogeneity in the welfare implications.

This paper presents three new findings. First, there are systematic changes in product entry, exit, and adjustments in market coverage over the business cycle, with the interaction between these dynamics and demand structure playing a critical role. Product markets with relatively less price-elastic demand—measured by the median within each industry—experienced significantly larger declines in entry during the Great Recession.² This finding is reinforced by the estimated higher sunk entry costs in less price-elastic markets, implying greater barriers for new entrants during recessions. Exit and adjustment also vary systematically, with effects concentrated among low-appeal products (more exits and larger reductions in the number of stores). The evidence from both the extensive and intensive margins is more pronounced when comparing responses across elasticities: products with less price-elastic demand are more likely to exit or reduce their market coverage. Overall, high-appeal products with inelastic demand remain relatively resilient, whereas low-appeal niche products are disproportionately vulnerable to exit. These findings indicate that

¹Many papers refer to this demand shifter as “quality,” but I use the term “appeal,” following Hottman, Redding, and Weinstein (2016), to remain agnostic about the source of demand variation, which may reflect vertical quality, subjective taste, or other unobserved factors beyond price and observed characteristics. This concept of product appeal also differs from productivity, which typically refers to a firm’s ability to produce more output given the same inputs.

²Because data on potential entrants are not available and ex ante heterogeneity cannot be observed, the entry analysis is limited to comparisons over time and across industries.

the interaction between elasticity and product appeal amplifies the product market dynamics during recessions.

Second, the welfare costs of business cycle driven by aggregate demand shocks are substantial. The estimated welfare loss amounts to approximately \$37 billion, or about 1.5% of pre-recession personal consumption expenditure on food products. These losses are concentrated in consumer surplus, as the aggregate demand shock lowers consumers' utility when remaining high-appeal products charge higher prices. This occurs because consumers are more sensitive to price increases than to gains in product appeal. The decline is significantly larger in less price-elastic markets, where limited substitution forces consumers to bear higher prices. Considering that consumer surplus accounts for roughly 10% of total social welfare, this result highlights that consumers are significantly affected by recessions through changes in product market dynamics.

Lastly, the counterfactual exercise reveals that accounting for demand heterogeneity in relation to product market dynamics is crucial for correctly quantifying the welfare costs of the business cycle. In the scenario without consumer heterogeneity, the demand system collapses to a logit structure, which closely approximates a constant elasticity of substitution (CES) demand system (Anderson, De Palma, and Thisse, 1992). In this case, the welfare costs are understated by about 5%. This understatement is substantial, as it amounts to roughly 0.2% of personal consumption expenditure on food products. Thus, this paper highlights that the interaction between demand heterogeneity and producers' dynamic decisions is essential for evaluating how recessions shape product spaces.

3

My framework relies on the NielsenIQ Retail Scanner Data, highly disaggregated product-level data from the U.S. consumer goods sector, which provide rich information on weekly sales for each product across local retail stores. The dataset includes detailed sales information, such as product characteristics, prices, and quantities sold. Using these data, I recover the demand system and construct a product-level state variable; product appeal. In addition, the dataset's broad coverage allows me to track product entry, exit, and market coverage adjustments over time. Utilizing this information, I identify how products with specific characteristics make dynamic decisions regarding

³Throughout the paper, I use the terms *industry* and *market* interchangeably. An industry is defined as the market for a specific good (e.g., cereal, yogurt, bottled water), which consists of a set of products.

entry, exit, and changes in store coverage, which I use in the dynamic estimation.

The granular information in the data disciplines my model of product market dynamics, which is designed to reflect the patterns observed in the data on both the consumer and producer sides. On the consumer side, the demand system incorporates heterogeneity in price sensitivity. Consumers also value product availability, which corresponds to producers' predetermined market coverage decisions, measured by the number of stores in which their products are sold. Finally, the demand system includes unobserved demand shifters as a state variable. On the producer side, the model incorporates heterogeneous and strategic producers that make dynamic decisions regarding entry, exit, and market coverage adjustments over time. Producers operate under aggregate demand shocks, which affect the overall market size each firm faces. They interact strategically through aggregate moments rather than fully tracking the entire distribution of competitors' states.

My analysis includes the estimation of the product market dynamics model, which is conducted in two separate steps. First, I estimate a random-coefficients logit demand system, following [Berry, Levinsohn, and Pakes \(1995\)](#). Because estimating this model is computationally intensive, I adopt the approximation technique developed by [Salanié and Wolak \(2022\)](#), which allows me to extend the estimation to multiple industries. I apply this procedure separately to each industry and use instrumental variables introduced by [Gandhi and Houde \(2019\)](#) to identify the demand parameters. The key measure I employ in the comparative analysis is the price elasticity of demand, which is recovered from the estimated parameters within the demand estimation.⁴

Second, I estimate the cost parameters governing product entry, exit, and adjustment in market coverage decisions over the business cycle. To do so, I first recover the state variables for each product: product appeal and the number of stores where producers currently sell their products. Based on these state variables, I then estimate a full-solution maximum-likelihood model ([Rust, 1987](#); [Igami, 2017](#)). The idea is to compare model-implied policies to observed behavior in the microdata, summarized by state variables and characteristics. Taking a full-solution approach is advantageous because the counterfactual analysis ultimately requires solving the model.

⁴In a simple monopoly setting, the price elasticity of demand is inversely related to market power through the Lerner index. However, relying on elasticity alone can be misleading in more complex environments.

Related Literature. A large body of literature on business cycle dynamics has emphasized the role of entry and exit at the firm, establishment, and product levels.⁵ Bilbiie, Ghironi, and Melitz (2012) demonstrate how cyclical entry of product varieties can amplify business cycle fluctuations through love-of-variety preferences. Jaimovich and Floetotto (2008) highlight supply-side effects, focusing on changes in the number of competitors and countercyclical markups. Etro and Colciago (2010) show that strategic interactions in oligopolistic markets play a key role in magnifying business cycle dynamics through changes in product variety. Over the past decades, this literature has advanced our understanding of how entry and exit shape aggregate fluctuations (Siemer, 2014; Lee and Mukoyama, 2015; Clementi and Palazzo, 2016; Hamano and Zanetti, 2017; Gamber, 2023). However, micro-level evidence on product dynamics and their interaction with demand structure remains limited, leaving important gaps in our understanding of how aggregate shocks reshape the product space through demand. This paper addresses this gap by providing new evidence and deriving aggregate implications that account for heterogeneity in demand and market structure.

This paper is closely related to the literature highlighting how the demand structure shapes macroeconomic outcomes over the business cycle. Bils and Klenow (1998) show that cyclical shifts in expenditure on luxuries and durables provide key insights into business cycle dynamics, while Orchard (2022) finds that demand shifts toward necessities during recessions disproportionately burden low-income households by contributing to price increases. Related studies emphasize how consumers reallocate spending across products in response to income changes. Jaimovich, Rebelo, and Wong (2019) document that recessions trigger a shift toward lower-quality, less labor-intensive goods, helping to explain cyclical movements in aggregate activity, and Argente and Lee (2021) show similar quality substitution and shopping adjustments during the Great Recession, which contribute to cost-of-living inequality. Michelacci, Paciello, and Pozzi (2022) further demonstrate that changes in consumption baskets play a central role in shaping aggregate demand. Building on these insights, this paper incorporates a demand system disciplined by microdata and links it to producers' entry and exit decisions. By jointly modeling consumer behavior and producer dynamics, the analysis explores how macroeconomic shocks generate heterogeneous responses across markets and products through the demand channel.

⁵In cases involving multi-product firms, adjustments in entry and exit often take the form of changes in product scope (Minniti and Turino, 2013; Pavlov and Weder, 2017, 2022; Pavlov, 2021).

Lastly, this paper connects to the literature on endogenous market structure. This area has been extensively studied in industrial organization, focusing on how the introduction of new products influences market competition and welfare, as shown in studies of the cereal market (Nevo, 2001) and the commercial vehicle market (Wollmann, 2018). In the context of dynamic entry and exit, Dunne et al. (2013) examine how these decisions shape the evolution of market structure, using the dentist industry as an example. Related studies incorporate aggregate demand uncertainty into models of market structure: Collard-Wexler (2013) illustrate its effects on production decisions in the ready-mix concrete industry, while Jeon (2022) analyze how business cycle uncertainty affects investment decisions in shipbuilding. This paper contributes to this literature by explicitly modeling producers' dynamic decisions alongside consumer demand, using a broader range of industries. Moreover, it extends the analysis of market structure to macroeconomic implications, highlighting how industry dynamics interact with aggregate conditions.

Outline. This paper is structured as follows. Section 2 describes the data and presents motivating facts about the U.S. product market over the period surrounding the Great Recession. Section 3 introduces the model of product market dynamics. Section 4 outlines the estimation methods. Section 5 presents results derived from product-level microdata. Section 6 presents counterfactual analyses that quantify the welfare costs of the business cycle. Finally, Section 7 concludes the paper.

2 Motivating Facts

This section introduces the data and presents motivating facts that illustrate how business cycles generate heterogeneous impacts across product markets. Section 2.1 describes the micro-level data from the U.S. consumer goods sector used in the analysis. Section 2.2 documents stylized facts that highlight differences in macroeconomic dynamics at both the aggregate and disaggregated levels.

2.1 Data Description

This paper uses the NielsenIQ Retail Scanner dataset from the Kilts Center for Marketing Data Center to study how the business cycle shapes product market dynamics. The dataset records prices and quantities sold at the Universal Product Code (UPC), store, and week levels, covering billions of transactions from 30,000–50,000 retail stores across the United States. It captures more than half

of total sales in grocery and drug stores and about 30% of sales in mass merchandisers nationwide, providing a representative view of U.S. consumption.⁶ Because of its scale and granularity, the dataset has become widely used in macroeconomic research on business cycles, firm dynamics, inflation, fiscal policy, and household inequality, among other topics.⁷

NielsenIQ also provides the Consumer Panel dataset, which has been extensively used in related work. However, the Retail Scanner dataset records billions of transactions directly at the point of sale and offers far greater scale, whereas the Consumer Panel relies on a sample of households who self-report their purchases with more limited product coverage. This broader coverage entails a higher computational burden but provides substantially richer information on product market dynamics and market structure, making it particularly well suited for studying the business cycle and its macroeconomic implications.

The dataset begins in 2006, so this study focuses on the business cycle episode surrounding the Great Recession. Focusing on this single recessionary episode still provides valuable evidence, as the Great Recession was one of the most severe demand-driven recession—well captured by the current framework based on aggregate demand shocks. While the analysis could, in principle, be extended to the COVID-19 recession, identifying producers' dynamic decisions during that period is complicated by multiple confounding factors. Accordingly, the analysis uses data from 2006 to 2019 and follows the definitions of entry and exit established in previous studies (Broda and Weinstein, 2010; Argente, Lee, and Moreira, 2018; Granja and Moreira, 2023). Entry is defined as the first period in which a product is sold in any store after at least four consecutive quarters with no prior sales, and exit as the last period in which the product is sold, followed by four consecutive quarters with no transactions. This timing rule improves the reliability of identifying entry and exit events by minimizing the influence of seasonality.

Consistent with these definitions, dynamic model estimation begins in the fourth quarter of 2007 (the onset of the Great Recession), while pooled data from 2006 through 2019 are used for demand estimation. For computational tractability and to capture structural changes over the business cycle, the dynamic model is estimated separately for three phases: 2007Q4–2009Q2, 2009Q3–2012Q4, and 2013Q1–2016Q4. I assume that economic conditions prior to the Great

⁶See <https://www.chicagobooth.edu/research/kilts/research-data/nielseniq> for details.

⁷For a review of studies using scanner data, see Dubois, Griffith, and O'Connell (2022).

Recession and from 2017Q1 onward represent normal periods without aggregate demand shocks. All estimations are conducted separately by industry, defined as the set of products within NielsenIQ categories, under the assumption that producers' decisions are independent across industries and that consumers' choices do not account for products from other industries.⁸ Throughout the paper, I use the terms “industry” and “product market” interchangeably.

Estimating each industry separately substantially reduces the computational burden and aligns with the objective of uncovering heterogeneity both across and within product markets. Several restrictions are applied to construct a sample suitable for examining product market dynamics over time. First, the analysis is limited to grocery stores, drug stores, and mass merchandisers, which together yield the most representative coverage of nationwide sales in the data.⁹ Second, only balanced stores within these chains are included to avoid confounding effects from store entry and exit. This restriction also ensures that producers' market coverage decisions, one of the key dynamic choices, are made within a fixed set of stores in the United States. Finally, private-label products are included in the demand estimation, but their entry and exit are treated as exogenous.¹⁰

Throughout the paper, products are defined at the distinct brand level to capture firms' strategies of product differentiation, following [Nevo \(2001\)](#). Defining products at the brand level reflects the idea that differentiation in consumer goods markets primarily occurs across brands rather than among individual UPCs within a brand. Each UPC at the barcode level is consolidated into a standardized unit measure (e.g., ounces, liters) within each brand, accounting for variations in package sizes and multipack configurations.¹¹ For example, the brand “Diet Pepsi” is treated as a single product in the analysis, encompassing multiple UPCs that represent different sizes and multipack offerings, such as a 2-liter bottle or an 8-pack of 12-ounce cans. Prices and quantities for each product are therefore expressed as price per unit and total units sold. This aggregation distinguishes the present study from previous research that examined cyclical properties of product markets at the UPC level. All prices are deflated using the Core Consumer Price Index.

⁸For instance, categories such as “soft drinks-carbonated” and “soft drinks-low calorie” are not combined, as NielsenIQ does not group them together. This assumption simplifies estimation but does not capture potential substitution across categories.

⁹Other outlets, such as convenience or liquor stores, are not representative in the data.

¹⁰NielsenIQ restricts information on private-label items to protect retailer identity, making it difficult to track their entry and exit.

¹¹Appendix Figure 1 shows that most brands consist of only a few unique UPCs, suggesting that the assumption of a single product at the brand level is not restrictive.

I further aggregate the data to higher levels to facilitate a more practical analysis, consistent with previous research (Nevo, 2001; Döpper et al., 2022). Weekly observations are aggregated to a quarterly frequency, which is standard in the business cycle literature. This quarterly aggregation mitigates potential biases from consumers’ dynamic demand behavior and reflects firms’ realistic decision-making horizon for product entry and exit.¹² Furthermore, sales data from individual stores are aggregated at the retail-chain level, so the uniform-pricing property across national retail chains does not affect the analysis (DellaVigna and Gentzkow, 2019). Regional aggregation is also performed at Nielsen’s Designated Market Areas (DMAs), which group together counties that share local television advertising markets.¹³

2.2 Heterogeneous Product Market Dynamics

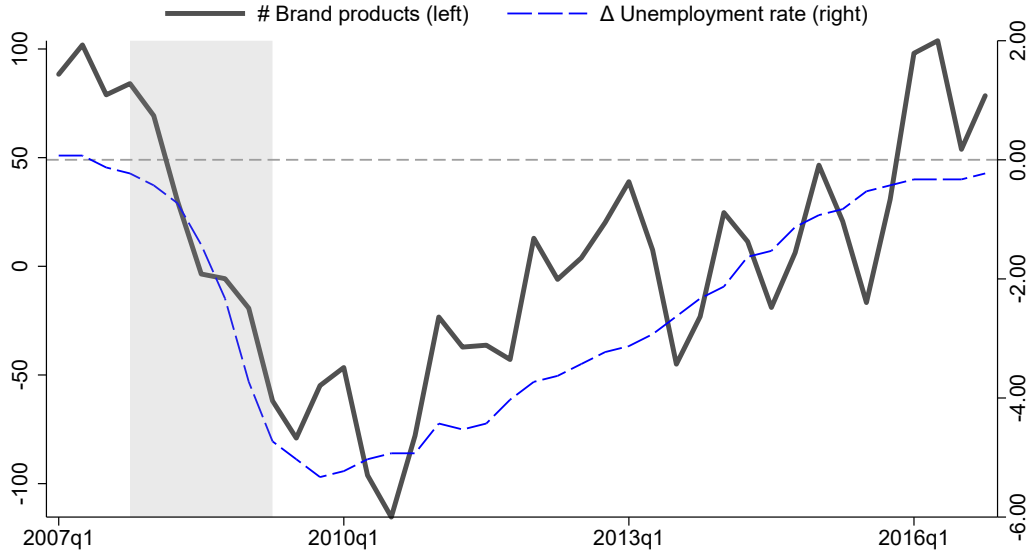
This paper is motivated by the fact that product market dynamics are substantially heterogeneous across industries, although aggregate patterns may appear similar. Figure 1 illustrates how this aggregate pattern can mask such heterogeneity. The number of incumbent product combinations across industries in the sample closely comoves with the normalized unemployment rate, dropping sharply during the Great Recession and recovering at a similar pace as the aggregate economy. The correlation between the two series is striking (0.85). This procyclical behavior is consistent with prior evidence that product entry and exit are highly sensitive to macroeconomic conditions (Broda and Weinstein, 2010; Argente, Lee, and Moreira, 2018). In particular, fluctuations in the total number of products are largely driven by changes in entry and exit dynamics.

Examining these dynamics at the industry level reveals substantial heterogeneity, suggesting that the economic consequences of aggregate shocks may diverge across sectors. Figure 2 plots this variation by showing the nine product categories with the largest sales shares in the sample. The black lines illustrate the number of incumbent products in each category, normalized to their 2007Q1 levels. The figure shows that some industries exhibit strong procyclical patterns, whereas others appear largely unaffected. Moreover, the magnitude and timing of fluctuations vary. For instance, industries such as cereal, canned soup, and tooth cleaners experience sharp declines in product variety during the recession, whereas chocolate, yogurt, and deodorants display relatively

¹²For example, consumers are unlikely to stockpile cereals three months in advance, and producers do not adjust their entry or exit decisions on a weekly basis.

¹³See <https://www.nielsen.com/dma-regions/>.

Figure 1: Aggregate Number of Incumbent Products over Time



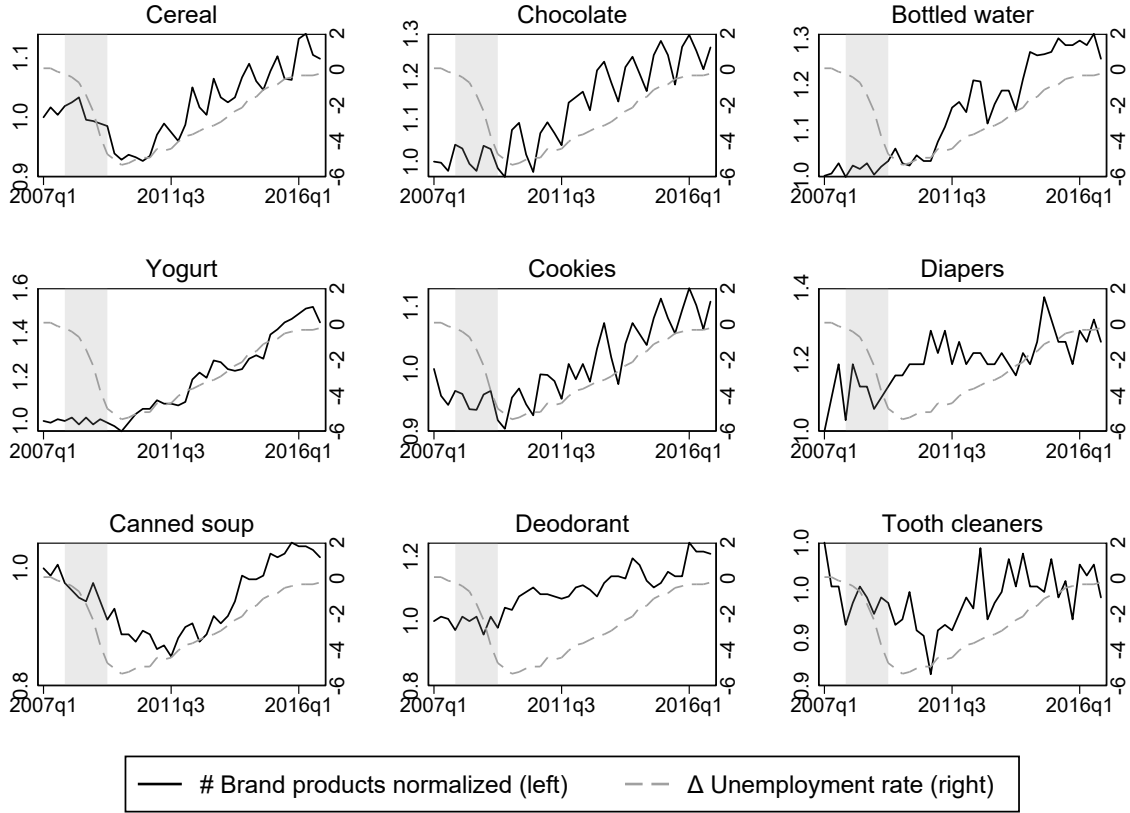
Notes: This figure plots the time series of detrended incumbent products and normalized unemployment rates. The solid black line (left axis) shows the number of incumbent products across industries in the NielsenIQ sample used for analysis. The dashed blue line (right axis) represents the aggregate economy, measured by the normalized unemployment rate, defined as the deviation from its pre-recession mean. The number of products is detrended using the Hodrick–Prescott filter after applying a three-quarter moving average. The shaded area indicates the NBER-defined recession period.

stable trends. These differences indicate that recessions do not uniformly affect product markets, underscoring the importance of understanding the mechanisms behind this heterogeneity.

Finally, Figure 3 further highlights this heterogeneity by comparing the impact of aggregate shocks across industries. It reports the change in the average number of products between the pre-recession and post-recession periods for the nine industries that experienced the largest declines in the sample. For example, pre-shave cosmetics (such as shaving cream) experienced a decline of about 25% in the number of products, whereas cookies fell by around 6.5%. Some of the industries that appeared strongly affected in the previous figure are not included here. Thus, comparing only pre- and post-recession averages would overlook substantial within-period variation in product market dynamics, underscoring the importance of capturing these dynamics in the analysis.

This paper argues that the heterogeneous responses of product markets to aggregate demand shocks arise from differences in demand structures within and across industries. Explaining them requires recovering the underlying demand parameters. Moreover, to fully capture product market dynamics, it is essential to account for producers' dynamic decisions: entry, exit, and adjustments

Figure 2: Number of Incumbent Products across Industries



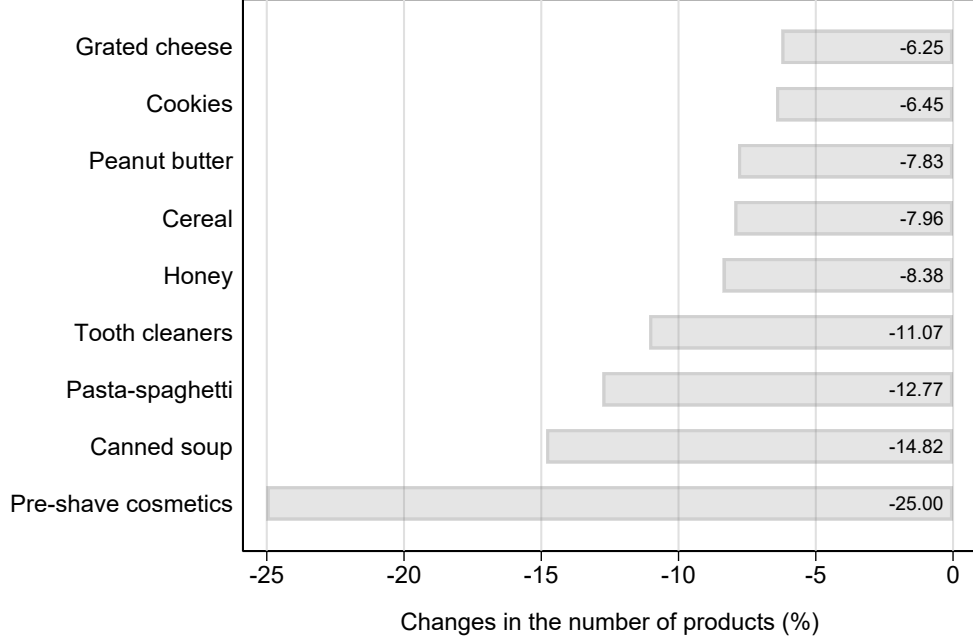
Notes: This figure plots the time series of the number of incumbent products (solid black line, left axis) for each product category, normalized to its 2007Q1 level. The nine panels display the top nine product categories ranked by average category-level revenue in the analysis sample. The dashed gray line (right axis) shows the normalized unemployment rate. Shaded areas indicate NBER-defined recession periods.

in market coverage. Throughout the analysis, I show how heterogeneous demand structures shape these dynamics and, in turn, their macroeconomic implications.

3 Model

This section develops a dynamic model of the product market, following the moment-based Markov equilibrium framework of [Ifrach and Weintraub \(2017\)](#) in the spirit of [Ericson and Pakes \(1995\)](#), and introduces aggregate demand shocks to capture business cycles. The model also incorporates a discrete-choice demand system with consumer heterogeneity ([Berry, Levinsohn, and Pakes, 1995](#)). The subsections describe the model environment (Section 3.1), the moment-based strategies of producers (Section 3.2), and the optimization problems of producers (Section 3.3).

Figure 3: Changes in the Average Number of Product between Pre-recession and Post-recession



Notes: This figure plots the percentage change in the average number of incumbent products after the recession relative to the pre-recession period for each industry. The figure displays the nine categories that experienced the largest declines.

3.1 Environment

Time. Time is discrete and finite. A period corresponds to a quarter, indexed by $t = 1, 2, \dots, \bar{T}$.

Producers. In each period, there is a finite number of incumbent producers, denoted J_t . Each producer indexed by j is a single-product firm, so that product characteristics are sufficient to describe the producer.¹⁴ In addition, there are J_t^{pe} potential entrants, representing the maximum number of producers the market can accommodate.

Producer state. Incumbent producer heterogeneity is characterized by two key state variables: *product appeal*, ξ_{jt} , and *market coverage*, N_{jt} . In each period, the individual state of producer j is denoted by the pair $x_{jt} = (\xi_{jt}, N_{jt}) \in \mathcal{X}$, where $\mathcal{X}$ denotes the finite set of possible state combinations. Product appeal captures favorable shifts in demand conditional on price and other observed characteristics. It follows a first-order Markov process:

$$\xi_{jt+1} = \mathbb{E}[\xi_{jt+1} \mid \xi_{jt}] + \eta_{jt+1}, \quad (1)$$

¹⁴This simplification is made for practical reasons, as estimating entry and exit dynamics for heterogeneous products within multi-product firms is computationally infeasible and highly challenging (Benkard, 2004).

where η_{jt+1} is an idiosyncratic demand shock drawn from an i.i.d. standard normal distribution. Market coverage reflects a product's availability to consumers, measured by the number of local retail stores in which it is sold.¹⁵ It is determined by producers each period and evolves over time through costly endogenous adjustment decisions. Potential entrants are ex ante homogeneous; their heterogeneity is realized upon entry as new producers.

Aggregate state. The aggregate state of the market is characterized by z_t and $\hat{s}_t$. The aggregate demand shock $z_t \in \mathcal{Z}$ represents the current level of market demand common to all producers. It captures the mass of consumers actively purchasing in the market; adverse shocks reduce this mass, shrinking the effective market size faced by producers. The aggregate demand shock follows a first-order Markov process:

$$z_{t+1} = \mathbb{E}[z_{t+1} \mid z_t] + \zeta_{t+1}, \quad (2)$$

where ζ_{t+1} is a white-noise shock. The variable $\hat{s}_t \in \mathcal{S}$ denotes the aggregate utility index, which summarizes the attractiveness of all products available in period t . Producers condition on this aggregate moment because it provides information about their product j 's profitability relative to industry-level demand. A detailed definition of $\hat{s}_t$ is provided in Section 3.2.

Exit process. In each period, incumbent producer j decides whether to remain in the market or exit. The exit decision is represented by the action a_{jt} , where $a_{jt} = 0$ indicates exit. If the producer exits, they receive a private exit value γ_{jt}^{ex} drawn from an exponential distribution with mean γ^{ex} .

Adjustment process. If the producer decides to stay, they choose the level of next period's market coverage, N_{jt+1} . The action of a staying producer is denoted by $a_{jt} = N_{jt+1}$ and evolves according to the following endogenous adjustment process:

$$N_{jt+1} = N_{jt} + \Delta_N, \quad (3)$$

where Δ_N represents the change in the number of stores in which producer j sells its product. The realization of next period's market coverage is subject to an exogenous noise term ν_{jt} , drawn from an i.i.d. Type I extreme value distribution with scale parameter ψ . Adjusting market coverage away from its current level incurs an adjustment cost.

¹⁵The total number of stores in the economy is fixed over time.

Entry process. In each period, each potential entrant decides whether to enter the market in the next period. The decision is denoted by a_{jt}^{pe} , where $a_{jt}^{pe} = 1$ indicates entry and $a_{jt}^{pe} = 0$ otherwise. If the entrant decides to enter, they incur a one-time private entry cost γ_{jt}^{en} drawn from an exponential distribution with mean γ^{en} . After paying the entry cost, the entrant's initial state is realized. All new entrants start with the lowest level of market coverage, $N_{jt+1}^{en} = 1$, and their product appeal ξ_{jt+1}^{en} is drawn from a time-invariant distribution G .

Consumer problem. In each period t , there are M_t consumers, indexed by i . Consumers are myopic and maximize static utility by choosing the product j that provides the highest utility, given price p_{jt} , market coverage N_{jt} , and product appeal ξ_{jt} . Each consumer is heterogeneous in price sensitivity, denoted by α_i , which follows a normal distribution with mean α and variance σ_α^2 . In addition, consumer i 's idiosyncratic taste for product j , denoted by ϵ_{ijt} , follows an i.i.d. Type I extreme value distribution.

If consumer i chooses product j among the J_t available products, the choice-specific indirect utility is given by

$$v_{ijt} = \alpha_i p_{jt} + \phi \log N_{jt} + \xi_{jt} + \epsilon_{ijt}. \quad (4)$$

The probability that consumer i chooses product j equals the probability that product j yields the highest utility among all available products, including the outside option with zero utility. This probability defines consumer i 's demand for product j , denoted d_{ijt} . Using the statistical property of ϵ_{ijt} under the Type I extreme value assumption, we can express d_{ijt} as

$$d_{ijt} = \frac{\exp[\alpha_i p_{jt} + \phi \log N_{jt} + \xi_{jt}]}{1 + \sum_{k \in J_t} \exp[\alpha_i p_{kt} + \phi \log N_{kt} + \xi_{kt}]}, \quad (5)$$

where the utility of the outside option is normalized to zero.

3.2 Moment-based strategies

Demand for product j depends not only on its own state but also on the states of all other producers. Equation (5) illustrates this interdependence: d_{ijt} is a function not only of $x_{jt} = (\xi_{jt}, N_{jt})$ but also of $x_{kt} = (\xi_{kt}, N_{kt})$ for $k \neq j$. However, it is difficult for producers to track the states of all competitors over time, as they evolve each period. Thus, producers summarize this information

using an aggregate moment that serves as a sufficient statistic for the distribution of competitors' states rather than tracking individual competitors' states over time (Ifrach and Weintraub, 2017).

The aggregate moment in this framework is the aggregate utility index $\hat{s}_t$, defined as

$$\hat{s}_t = \sum_{x \in \mathcal{X}} \mu_t(x) \exp [\alpha_i p_t(x) + \phi \log N_t(x) + \xi_t(x)]. \quad (6)$$

Let μ_t denote the distribution of producers across states. Then $\mu_t(x)$ represents the number of producers in state $x \in \mathcal{X}$, while $\xi_t(x)$ and $N_t(x)$ denote the levels of product appeal and market coverage associated with that state, respectively. This index summarizes the overall valuation of all products in the market and, hence, affects each producer's own demand—for instance, when this valuation rises, other things equal, the relative utility of product j declines. For each producer j , this aggregate moment provides a key signal about its own profitability, on which producers rely when making dynamic decisions.

Now, I define the moment-based strategies for each producer as functions of the moment-based state s_t , consistent with the assumption that strategies depend on a few sufficient statistics of the industry state rather than the entire distribution. Let λ_t denote the profile of all producers' actions, $\lambda_t = \{a_{jt}\}_{j \in J_t} \cup \{a_{jt}^{pe}\}_{j \in J_t^{pe}}$, which depends on individual states, the aggregate state, and idiosyncratic shocks. Given λ_t , which summarizes all producers' strategies at time t , the distribution of producers μ_t evolves according to

$$\mu_{t+1}(x') = \begin{cases} \sum_{x \in \mathcal{X}} Q_t(x' | x, \lambda_t) \mu_t(x) + \left(\sum_{j \in J_t^{pe}} a_{jt}^{pe} \right) dG(\xi') & \text{if } N' = N^{en}, \\ \sum_{x \in \mathcal{X}} Q_t(x' | x, \lambda_t) \mu_t(x) & \text{otherwise,} \end{cases} \quad (7)$$

where $x' = (\xi', N')$ and $Q_t(x' | x, \lambda_t)$ denotes the transition kernel governing the evolution of incumbents' states conditional on the strategy profile. In what follows, I assume that the aggregate utility index s_t evolves approximately as a Markov process.

3.3 Dynamic Optimization

At the beginning of period t , the individual states of all incumbent producers, $x_{jt} = (\xi_{jt}, N_{jt})$, the aggregate demand shock z_t , and the aggregate industry state $\hat{s}_t$ are realized. Individual demand can

be expressed as $d_{ijt}(\xi_{jt}, N_{jt}, \hat{s}_t; \alpha, \sigma_\alpha^2, \phi)$, which depends on the producer's own state, the market state, and the demand parameters. Combining the individual and aggregate components, I denote the overall state at time t as $s_{jt} = (\xi_{jt}, N_{jt}, \hat{s}_t, z_t)$ for notational simplicity. Noting that the aggregate demand shock z_t scales the overall market size, I obtain the aggregate quantity demanded for product j by integrating over the distribution of consumer price sensitivities:

$$D_{jt}(s_{jt}; \alpha, \sigma_\alpha^2, \phi) = \left(\int d_{ijt}(\xi_{jt}, N_{jt}, \hat{s}_t; \alpha, \sigma_\alpha^2, \phi) dF(\alpha_i) \right) M_t \exp(z_t). \quad (8)$$

In each period, an incumbent producer earns static profit

$$\pi_{jt}(s_{jt}; \alpha, \sigma_\alpha^2, \phi) = [p_{jt}^*(\xi_{jt}, N_{jt}) - mc_{jt}(s_{jt}; \alpha, \sigma_\alpha^2, \phi)] D_{jt}(s_{jt}; \alpha, \sigma_\alpha^2, \phi), \quad (9)$$

where $p_{jt}^*(\xi_{jt}, N_{jt})$ is the observed equilibrium price expressed as a function of individual states that evolve over time, and $mc_{jt}(s_{jt}; \alpha, \sigma_\alpha^2, \phi)$ is the marginal cost. To maintain tractability, I do not explicitly model pricing decisions within the dynamic framework. Instead, this model assumes that producers compete in prices according to a static Bertrand–Nash pricing game in each period t , and that the corresponding first-order condition holds. Within the dynamic model, the marginal cost $mc_{jt}(s_{jt}; \alpha, \sigma_\alpha^2, \phi)$ is recovered from this static first-order condition using observed prices and the estimated demand system.¹⁶

After earning period profit, incumbents observe their private shock components related to exit, γ_{jt}^{ex} , and market coverage adjustment, ν_{jt} . Each incumbent then decides whether to stay in the market or exit. In making this decision, they consider the expected continuation value, discounted by $\beta \in (0, 1)$, and the action profile λ_t that summarizes other producers' strategies. The dynamic optimization problem of an incumbent is summarized by the following Bellman equation:

$$V_t(s_{jt}, \lambda_t; \theta) = \pi_t(s_{jt}; \alpha, \sigma_\alpha^2, \phi) + \mathbb{E}_{\gamma^{ex}} \left[\max \left\{ \gamma_{jt}^{ex}, \right. \right. \\ \left. \left. \mathbb{E}_\nu \left[\max_{N'} \left\{ -C(N', N_{jt}; \gamma^c) + \nu_{jt} + \beta \mathbb{E}_t [V_{t+1}(s_{jt+1}, \lambda_{t+1}; \theta) | s_{jt}, \lambda_t] \right\} \middle| s_{jt} \right] \right\} \middle| s_{jt} \right], \quad (10)$$

where $\theta = \{\alpha, \sigma_\alpha^2, \phi, \gamma^{en}, \gamma^{ex}, \gamma^c, \psi\}$, and $\gamma^c = (\gamma_0, \gamma_1^+, \gamma_1^-, \gamma_2)$. The adjustment cost function is

¹⁶Under the single-product assumption, the recovered marginal costs may be biased for multi-product firms.

defined as

$$C(N_{jt+1}, N_{jt}; \gamma^c) = \mathbb{1}\{N_{jt+1} > N_{jt}\} (\gamma_0 + \gamma_1^+ \Delta_N + \gamma_2 \Delta_N^2) \\ + \mathbb{1}\{N_{jt+1} < N_{jt}\} (\gamma_0 + \gamma_1^- \Delta_N + \gamma_2 \Delta_N^2), \quad (11)$$

where the linear adjustment parameters are asymmetrically specified.

At the same time, each potential entrant draws a private entry cost γ_{jt}^{en} . Entrants decide whether to enter the market by comparing this cost with their expected discounted value upon entry, given the current aggregate states and action profile, by solving the following problem:

$$\mathbb{E}_{\gamma^{en}} \left[\max \left\{ 0, \beta \mathbb{E}_t \left[V_{t+1}(s_{jt+1}, \lambda_{t+1}; \theta) \mid \hat{s}_t, z_t, \lambda_t \right] - \gamma_{jt}^{en} \right\} \mid \hat{s}_t, z_t \right]. \quad (12)$$

Producers that choose to enter become incumbents in period $t + 1$, with initial states determined by a product-appeal draw $\xi_{jt+1} \sim G(\cdot)$ and $N_{jt}^{en} = 1$. From period $t + 1$ onward, they solve the dynamic optimization problem as incumbents.

This paper adopts the moment-based Markov equilibrium framework of [Ifrach and Weintraub \(2017\)](#). To ensure the existence of an equilibrium and computational feasibility, I assume that the industry's nonstationary evolution converges to its stationary counterpart over a sufficiently long time horizon ([Benkard, Jeziorski, and Weintraub, 2024](#)). The dynamic problem is solved by backward induction, where the terminal period's value function is approximated as if no producers exit the market and each continues to earn profits indefinitely without adjusting market coverage, while only the product appeal ξ_{jt} evolves according to its transition probability.

In equilibrium, the moment-based strategy λ_t satisfies the following conditions for all periods t : incumbent producers solve (10), potential entrants solve (12), aggregate demand shocks evolve according to (2), the number of producers evolves according to (7), and each producer forms beliefs about the industry state based on the moment-based states in (6).

4 Estimation

This section describes a two-step estimation procedure. In Section 4.1, I estimate key demand parameters in a static framework following [Berry, Levinsohn, and Pakes \(1995\)](#), approximating

the demand equation using the method of [Salanié and Wolak \(2022\)](#) to handle multiple industries efficiently. In Section 4.2, I estimate the parameters governing producers' dynamic optimization by maximum likelihood using a nested fixed-point algorithm ([Rust, 1987](#)).

4.1 Demand Estimation

The demand structure of this paper is described in equation (4). Consumers are heterogeneous in their price sensitivity ([Berry, Levinsohn, and Pakes, 1995](#)). Each industry has its own demand parameters for the distribution $(\alpha, \sigma_\alpha^2)$ and the preference for product availability ϕ . To capture heterogeneity both within and across industries, I estimate the demand parameters separately for each industry. However, estimating random-coefficient logit demand models is computationally intensive even for a single industry, and extending this approach to many industries substantially increases the computational burden. To address this challenge, I follow the approach of [Salanié and Wolak \(2022\)](#), which approximates the nonlinear demand system with a linear equation. This method has also been applied to multiple product categories in recent work by [Brand \(2021\)](#), who show that results from both methods are qualitatively similar.

For the demand estimation, I use information on prices and quantities from local stores, defined as the combination of region and retail chain in each quarter. This level of aggregation is standard in the literature, as it exploits variation in demand across local markets. In addition, obtaining consistent estimates of demand parameters under the approximation requires a sufficiently large number of markets. For these reasons, I estimate the model at the local market level and assume that the preference parameters within each industry are identical and remain unchanged under any aggregation. To maximize the number of markets, I pool data across all quarters from 2006 to 2019 within each industry for estimation.

In this approximated estimation, the demand system can be estimated in a manner similar to the standard logit demand model. Following [Berry \(1994\)](#), we apply the logit inversion using the outside-option market share to equation (8). Since I estimate the demand system at the local market level, each market m 's aggregate demand is expressed for this inversion. Applying the

approximation method of [Salanié and Wolak \(2022\)](#), the logit inversion can be expressed as

$$\log \left(\frac{D_{jmt}}{D_{0mt}} \right) = \alpha p_{jmt} + \sigma_\alpha^2 K_{jmt} + \phi \log N_{jmt} + \xi_{jmt} + \mathcal{O}(\cdot), \quad (13)$$

where K_{jmt} is an *artificial regressor* defined as

$$K_{jmt} = \left(\frac{p_{jmt}}{2} - e_{mt} \right) p_{jmt}, \quad (14)$$

and

$$e_{mt} = \sum_{j=1}^{J_{mt}} D_{jmt} p_{jmt}, \quad (15)$$

which represents the prices in market m weighted by market shares.

Estimating equation (13) requires two instrumental variables (one for price and one for the artificial regressor, which contains both market shares and prices). I employ three instruments: a quadratic version of the differentiation IVs ([Gandhi and Houde, 2019](#)), constructed using prices and the number of stores in which product j is sold, and the local unemployment rate ([Backus, Conlon, and Sinkinson, 2021](#)). To construct the price instrument, I first create the [Hausman \(1994\)](#) instrument and then extract retailer-specific price changes by subtracting the national average, following [Allcott, Lockwood, and Taubinsky \(2019\)](#) and [Conlon and Gortmaker \(2023\)](#). I then recover predicted prices by regressing observed prices on the Hausman instrument with a set of fixed effects; these predicted prices are subsequently used to generate the differentiation IV.

For the empirical specification, I include product fixed effects and market fixed effects. The aggregate demand shocks are absorbed by these fixed effects, so the recovered demand parameters should be interpreted as conditional on these shocks. To calculate market shares, I define the market size as the total population multiplied by the number of store combinations (disaggregated) in each local market. I also normalize market shares so that the combined share of inside goods is approximately 0.45, following [Döpper et al. \(2022\)](#). The estimation sample for the demand system includes products with private labels. Because it is not possible to identify the manufacturers of these products, I treat their dynamics as exogenous and do not model their endogenous decisions; they are therefore excluded from the dynamic estimation sample. After estimation, I recover the product appeal ξ_{jmt} by adding the recovered product fixed effect and then aggregate both product

Table 1: Estimates of Demand Parameters

Parameters	Notation	Cereal	Yogurt
Price sensitivity (mean)	α	-14.95 (3.03)	-38.83 (5.46)
Price sensitivity (variance)	σ_α^2	15.71 (5.99)	71.79 (37.16)
Product availability	ϕ	0.91 (0.00)	0.93 (0.05)
Autocorrelation of product appeal		0.83	0.80
Median price elasticity of demand		-2.57	-4.50

Notes: Robust standard errors are in parentheses. Results for two industries commonly studied in the literature are presented as illustrative examples. The product-level price elasticity of demand is computed as $\frac{\partial D_{jt}}{\partial p_{jt}} \cdot \frac{p_{jt}}{D_{jt}}$. Median price elasticities are calculated within each industry as the median across all sample periods.

appeal and prices at the national level using quantity weights. Finally, I measure the number of stores as the count of chain $\times$ region combinations in which product j is sold each quarter.

Table 1 reports the estimated demand parameters and the recovered median price elasticities for the cereal and yogurt industries as illustrative examples. The current estimation sample covers 15 industries in total. The estimates align well with values documented in the literature, with the median price elasticity of ready-to-eat cereal calculated at -2.57 and yogurt at -4.50 .¹⁷ The results highlight two key points. First, there is notable within-industry heterogeneity, as captured by the positive and significant variance parameters. Consumers dislike higher prices, but the degree of this price sensitivity varies across individuals. Second, the demand structures differ significantly between the two industries. The mean price sensitivity in yogurt is more than twice that of cereal, and the variance of price sensitivity is also substantially larger. This indicates meaningful across-industry heterogeneity. In the following analysis, I examine how these differences in demand structures contribute to the heterogeneous responses of product market dynamics to aggregate shocks.

¹⁷Integration is performed using Gauss–Hermite quadrature of order 11.

4.2 Estimating Dynamic Model

This paper estimates the dynamic parameters using a full-solution approach with a nested fixed-point algorithm, following Rust (1987) and Igami (2017). I use maximum likelihood estimation to rationalize the observed product market behavior. For the state-space specification, product appeal is discretized into 11 grid points, and transition probabilities are computed directly from the data using Laplace smoothing. The number of stores is discretized into 20 grid points. Unlike product appeal—which is recovered as a residualized variable from the demand estimation—the number of stores is right-skewed with a long tail. To reduce the state-space dimension, I cap the number of stores at 200 and set the grid size to 10, starting from 1.¹⁸

Because product market dynamics during the Great Recession are highly nonlinear and vary substantially over time, estimating the model over the entire sample period (2007–2018) would not adequately capture these transitional patterns. The environment is nonstationary: aggregate demand and producers’ decisions evolve over time across distinct phases rather than remaining constant. Therefore, I estimate the model separately for three subperiods: 2007Q4–2009Q2, 2009Q3–2012Q4, and 2013Q1–2016Q4. For the pre-recession period and after 2016, I assume that the economy returns to normal conditions without aggregate demand shocks.¹⁹

The dynamic estimation aims to recover the parameters that govern producers’ entry and exit decisions, $(\gamma^{en}, \gamma^{ex})$, and market coverage adjustment decisions, $(\gamma_0, \gamma_1^+, \gamma_1^-, \gamma_2)$. The underlying idea is to identify the parameters that best rationalize the observed behavior of each product over time. This is achieved by computing the likelihood of the observed data conditional on the model-implied optimal policies, which characterize producers’ dynamic decision probabilities. The logit scale parameter ψ , which governs the stochasticity of market coverage adjustments, is estimated separately, after recovering the other dynamic parameters, using a minimum-distance procedure that matches the aggregate number of products over time to its model-implied counterpart.

¹⁸In the pooled sample across all industries, 95 percent of observations have fewer than 100 stores.

¹⁹In the current framework, I model the aggregate demand shock as following a deterministic trajectory proxied by the normalized unemployment rate to capture the time-varying environment during and after the Great Recession. This approach rationalizes observed transitions in product market behavior while maintaining computational tractability. However, it abstracts from stochastic aggregate shocks that could alter firms’ expectations and reoptimization decisions each period. Benkard, Jeziorski, and Weintraub (2024) develop a framework that explicitly incorporates such stochastic aggregate shocks within a nonstationary Markov equilibrium, showing that equilibrium strategies may deviate when aggregate uncertainty is high. Extending the current model in this direction is an important avenue for future research.

The conditional choice probabilities solved from the model for incumbents are denoted by $\Pr_t(a_{jt} | s_{jt}, \lambda_t, \theta)$, where a_{jt} summarizes the action of an incumbent. Similarly, the conditional choice probabilities for potential entrants are defined as $\Pr_t(a_{jt}^{pe} | s_{jt}, \lambda_t, \theta)$, with a_{jt}^{pe} summarizing the action of a potential entrant. The likelihood function is then computed as follows:

$$\mathcal{L}(a; \theta) = \prod_{t=1}^T \left[\prod_{j \in J_t} \Pr_t(a_{jt} | s_{jt}, \lambda_t, \theta) \prod_{j \in J_t^{pe}} \Pr_t(a_{jt}^{pe} | s_{jt}, \lambda_t, \theta) \right], \quad (16)$$

where a summarizes the action profile of all producers observed over the sample period.

The maximum likelihood estimates are obtained by solving

$$\hat{\theta} = \arg \max_{\theta} \ln [\mathcal{L}(a; \theta)]. \quad (17)$$

Table 2 reports the recovered parameters from the maximum likelihood estimation during the Great Recession. Because this estimation incorporates price and quantity information disciplined by the data, the estimated dynamic costs (values) are expressed in dollar units. Estimates for other sample periods are reported in the Appendix. The results indicate that several parameters differ noticeably between the two industries. In particular, exit values are significantly higher in the cereal industry (\$85.91 million) than in the yogurt industry (\$36.58 million). Adjustment costs also exhibit a slightly more asymmetric structure in the yogurt industry, where the linear cost of store expansion is approximately \$90,000, while the linear cost of store reduction is about $-\$76,000$. This suggests that producers in the yogurt market face substantial costs even when contracting their market coverage.

To assess how well the estimated dynamic parameters explain the observed product market behavior, I evaluate the model's fit by comparing key equilibrium outcomes implied by the model with those observed in the data. Table 3 reports summary statistics of dynamic behaviors, including the average share of new entrants among incumbents, the mean exit rate among incumbents, and the average change in the number of stores. The simulated moments closely match their empirical counterparts, indicating that the model successfully replicates producers' dynamic decisions in both industries. The close correspondence between data and simulations suggests that the estimated parameters governing entry, exit, and adjustment costs provide an internally consistent explanation

Table 2: Estimates of Dynamic Parameters

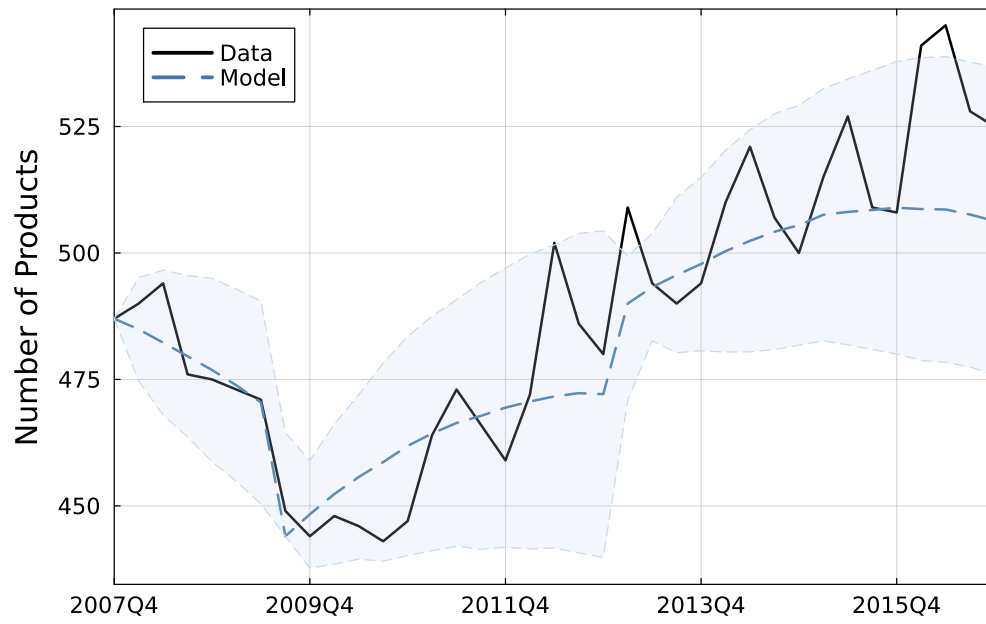
Parameters	Notation	Cereal	Yogurt
Mean entry cost (\$mn)	γ^{en}	855.16 (268.47)	882.63 (327.26)
Mean exit value (\$mn)	γ^{ex}	85.91 (27.71)	36.58 (12.96)
Fixed cost for store # change (\$mn)	γ_0	2.55 (0.06)	1.97 (0.16)
Linear cost for + store # change (\$k)	γ_1^+	55.71 (2.70)	90.31 (10.65)
Linear cost for – store # change (\$k)	γ_1^-	-9.57 (4.53)	-76.36 (21.05)
Convex cost for store # change	γ_2	1.73 (30.66)	0.91 (233.75)
Logit shock scale	ψ	1.01 (0.09)	1.15 (0.05)

Notes: Standard errors are in parentheses. Results for two industries commonly studied in the literature are presented as illustrative examples. The estimates are based on the sample period 2007Q4–2009Q2, which captures the Great Recession. Results for other sample periods are reported in the Appendix. All dollar values are expressed in 2006Q1 dollars.

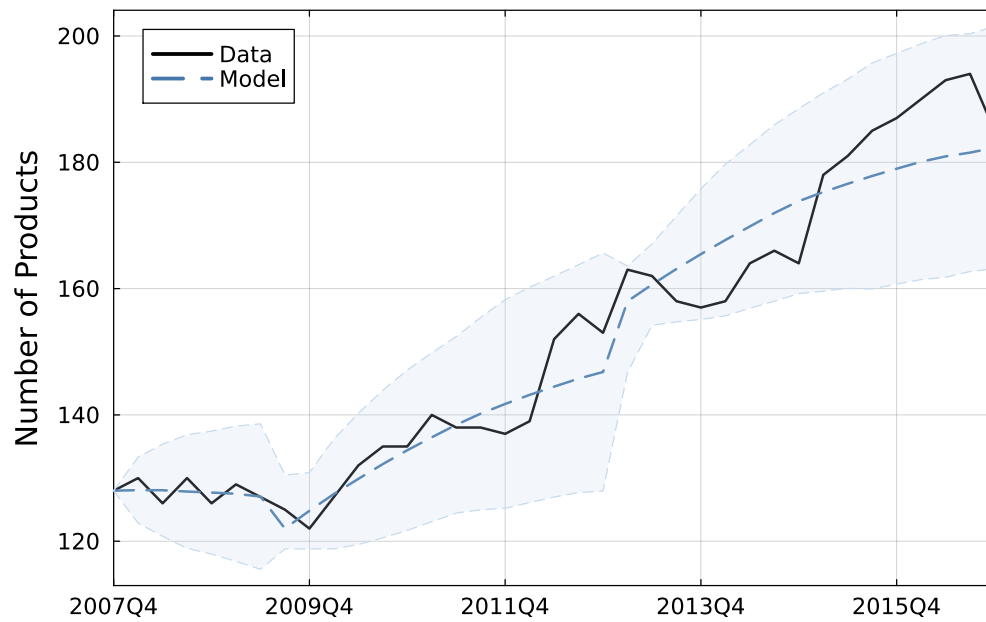
of producers' observed dynamic behavior at the product level.

Figure 4 compares the model-simulated aggregate number of products with the observed data for the two industries over the full nonstationary estimation sample (2007Q4–2016Q4). Figures for other industries are reported in the Appendix. The solid black line represents the data, while the dashed blue line shows the model's simulated path, with the shaded area denoting the 95% confidence intervals based on 500 simulations. The model successfully replicates both the level and the dynamics of product market activity. In the cereal industry (Panel A), the model captures the sharp contraction during the Great Recession and the subsequent persistent upward trend in product variety. In contrast, the yogurt industry (Panel B) exhibits a smoother decline over the same period, which the model also closely tracks. Overall, the fit suggests that the estimated dynamic parameters successfully explain the observed aggregate patterns and micro-level behaviors.

Figure 4: Model Fit from Aggregate Behavior



(A) Cereal



(B) Yogurt

Note: Standard errors for the model are calculated using 500 simulations. The model fit is based on the full nonstationary estimation sample period, 2007Q4–2016Q4. For each phase, the number of products in the first period is directly taken from the data. 95 percent confidence intervals are shown for the simulated series, with the first-period intervals recalculated using a moving average.

Table 3: Model Fit from Micro Behavior

	Data	Model
Panel A. Cereal		
Entrant share (%)	3.93	3.90 (0.00)
Exit rate (%)	3.59	3.77 (0.01)
Changes in the # of stores	3.87	3.72 (0.01)
Panel B. Yogurt		
Entrant share (%)	4.38	4.36 (0.01)
Exit rate (%)	3.31	3.46 (0.01)
Changes in the # of stores	2.83	2.79 (0.01)

Notes: Standard errors for the model are calculated using 500 simulations. The model fit is based on the sample period 2007Q4–2009Q2.

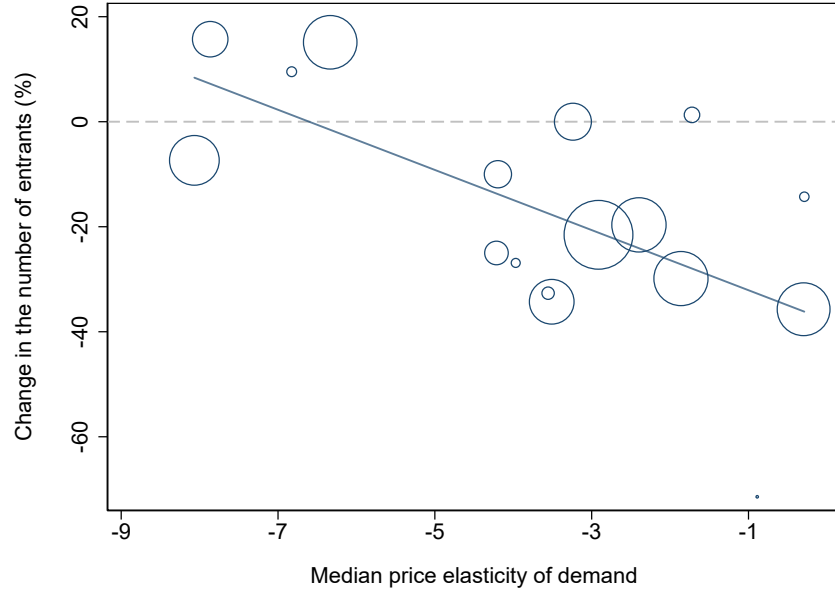
5 Implications of Estimated Model

This section presents empirical patterns on how business cycles affect product market dynamics, emphasizing their connection to demand heterogeneity implied by the estimated model. I relate observed entry, exit, and adjustment behaviors to model-based measures of product appeal and price elasticity of demand. The first subsection examines differences across industries (Section 5.1), while the second focuses on within-industry heterogeneity (Section 5.2).

5.1 Demand Heterogeneity across Industries

Figure 5 illustrates how demand heterogeneity, measured by the median price elasticity of demand in each industry, shapes entry responses over the business cycle. Each dot represents an industry, where the x-axis reports the median price elasticity of demand (derived from the model estimates) and the y-axis shows the percentage change in the number of entrants during the recession relative to the pre-recession period. Industries with less elastic demand (toward the right, with smaller

Figure 5: Changes in Product Entry across Industries



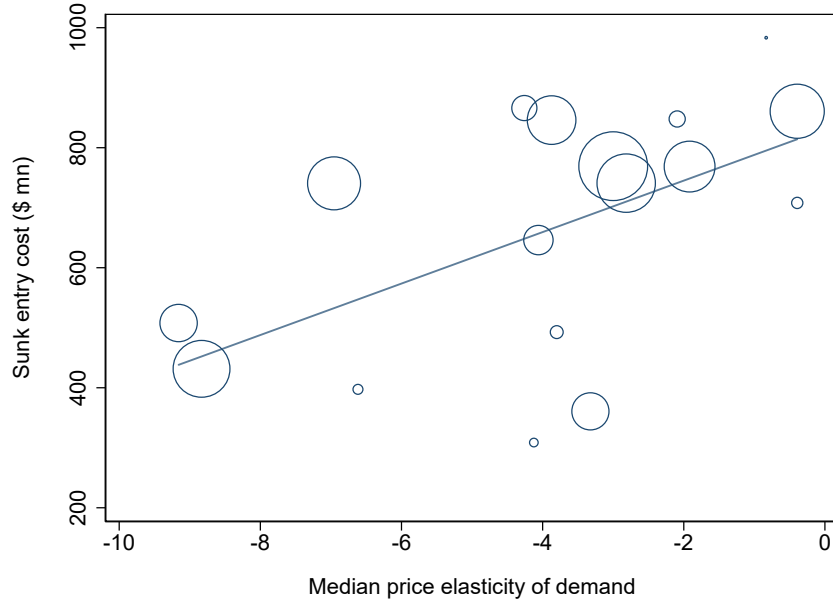
Note: Each dot represents the percentage change in the number of entrants across industries during the Great Recession compared to pre-recession. Dot size reflects total industry revenue.

absolute elasticities) experienced a larger decline in entry—up to a 40% reduction—as indicated by the downward-sloping relationship. The intuition is straightforward: when consumers are less sensitive to price changes, entrants face greater difficulty attracting demand even when offering lower prices, as consumers are less willing to substitute toward new products. This challenge is amplified during downturns, when overall market size contracts, leading to a sharper decline in entry.

This relationship can also be explained by the estimated dynamic costs. Figure 6 shows that industries with less elastic demand tend to have higher sunk entry costs: almost twice as large as those in more elastic industries. I calculate the industry-level sunk entry cost as the difference between the mean entry cost and the mean exit value, i.e., $\gamma^{en} - \gamma^{ex}$. Each dot represents an industry, where the x-axis plots the median price elasticity of demand and the y-axis shows the corresponding sunk entry cost. The positive relationship indicates that when consumers are less sensitive to price changes, producers are willing to incur larger upfront costs to enter the market. However, these higher sunk costs, in turn, make entry more difficult and help explain why industries with less elastic demand experience sharper declines in entry during recessions.

Lastly, Figure 7 plots changes in the number of exits during the Great Recession relative to the

Figure 6: Sunk Entry Costs across Industries



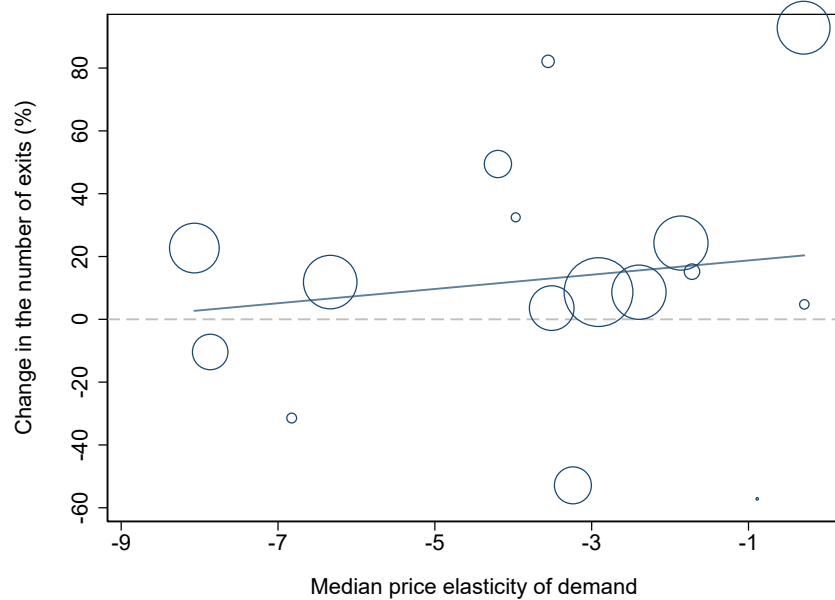
Note: Each dot represents the estimated sunk entry cost for an industry during the Great Recession. Dot size reflects total industry revenue.

pre-recession period. Unlike entry, the relationship between exit and demand heterogeneity is not as strong. While industries with less elastic demand tend to experience slightly larger increases in exits, the overall pattern remains weak. Nonetheless, some industries exhibit substantially higher exit rates than others. On average, exits increased by about 12% across industries, suggesting that additional factors beyond median price elasticity explain exit behavior. Moreover, exit decisions exhibit greater within-industry variation, as they depend on product-specific characteristics (e.g., product appeal) and demand elasticities. This contrasts with entry decisions, which are made by ex-ante homogeneous potential entrants. In the next subsection, I explore within-industry heterogeneity in incumbents' dynamic behavior in greater detail.

5.2 Product Heterogeneity within Industry

Figure 8 summarizes how incumbents adjust their dynamic behavior during recessions. Panel (A) shows that exit rates rise sharply by about 0.7 percentage points for products with below-median appeal, indicating that niche products are more vulnerable during downturns. In contrast, high-appeal products remain relatively stable throughout the period. Panel (B) again illustrates

Figure 7: Changes in Product Exit across Industries

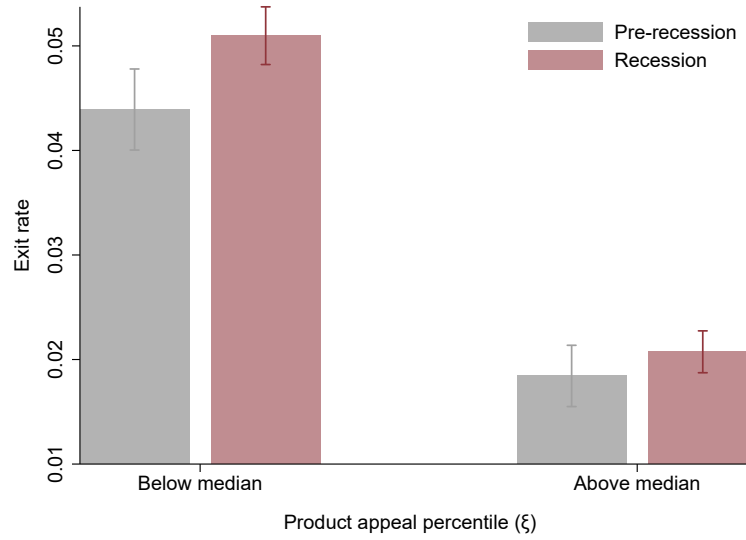


Note: Each dot represents the percentage change in the number of exits across industries during the Great Recession compared to pre-recession. Dot size reflects total industry revenue.

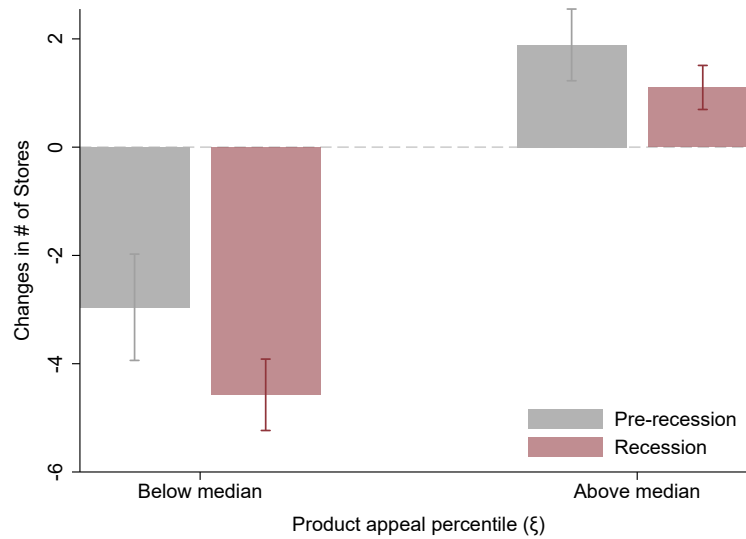
this asymmetric behavior by showing changes in market coverage, measured by the number of stores carrying each product. Low-appeal products generally reduce their coverage by about three stores, whereas high-appeal products expand it by around two on average. The recession amplifies this divergence: low-appeal products experience a substantial contraction of roughly five stores, while high-appeal products largely sustain their presence. Together, these behaviors imply that recessions intensify selection among incumbents, disproportionately affecting weaker products.

Figure 9 examines how another aspect of demand structure further shapes producers' dynamic decisions during recessions. The figure divides products into elastic and less-elastic groups and compares exit rates (Panel A) and changes in market coverage (Panel B) by product appeal. Within both groups, low-appeal products are more likely to exit, but the effect is particularly pronounced among less-elastic products by about one percentage point relative to low-appeal products with elastic demand. The contrast is even stronger in market coverage: low-appeal products in less-elastic industries reduce their coverage by an average of seven stores, whereas products with more elastic demand or higher appeal exhibit little change. The intuition behind this pattern is that when low-appeal producers face less-elastic demand, cutting prices does little to attract new consumers, as they are less responsive to price changes. Consequently, these products

Figure 8: Changes in Exit and Market Coverage Adjustment



(A) Exit



(B) Market Coverage Adjustment

Note: Each bar represents the mean exit rate (Panel A) and the average change in the number of stores carrying each product (Panel B) before and during the Great Recession. Products are divided into below- and above-median groups based on recovered product appeal (ξ). The pre-recession period covers 2007Q1–2007Q3, and the recession period covers 2007Q4–2009Q2. Error bars indicate 90% confidence intervals.

are more likely to exit or substantially reduce their market presence during recessions. These results suggest that demand elasticity and product appeal jointly determine producers' dynamic behavior during recessions.

6 Counterfactual Analysis

So far, the analysis has shown that product market dynamics vary substantially across industries, with demand heterogeneity serving as a key driver of these differences. This section conducts counterfactual experiments to quantify the welfare implications of these heterogeneous dynamics. I first examine how business cycle fluctuations affect welfare by measuring the welfare costs of the business cycle (Section 6.1). Next, I evaluate how these welfare costs differ across industries (Section 6.2). Finally, I conduct a counterfactual experiment that removes demand heterogeneity (Section 6.3).

6.1 Welfare Costs of the Business Cycle

First, I compute consumer and producer surplus at time t using the estimated parameters from the previous section. Consumer surplus, CS_t , is defined as

$$CS_t = \int \left\{ \frac{\gamma_E + \ln \left[1 + \sum_{j \in J_t} \exp(\alpha_i p_{jt} + \phi \log N_{jt} + \xi_{jt}) \right]}{-\alpha_i} \right\} dF(\alpha_i) \cdot M_t \cdot \exp(z_t), \quad (18)$$

where γ_E denotes the Euler constant (≈ 0.57721). The term divided by $-\alpha_i$ converts utility units into monetary value, consistent with the inclusive value formulation.

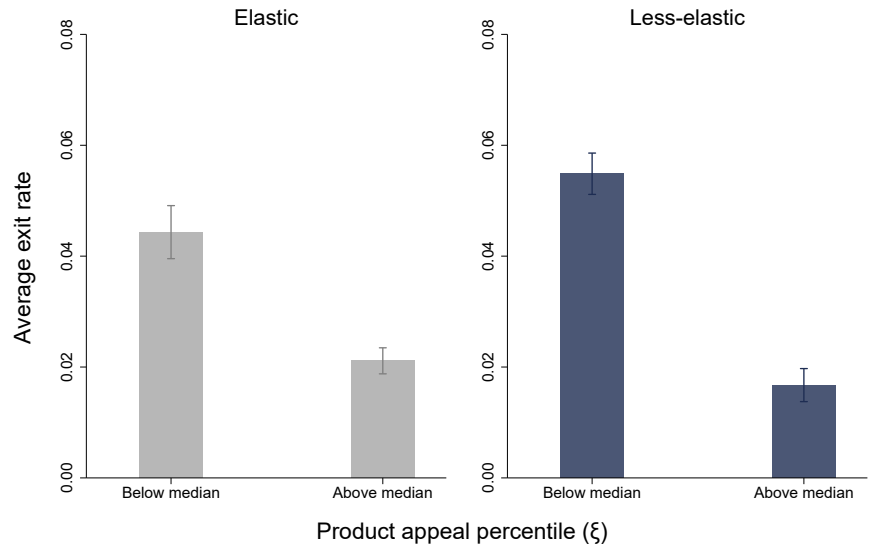
Similarly, producer surplus, PS_t , is defined as

$$PS_t = \sum_{j \in J_t} \left\{ \pi_t(s_{jt}; \alpha, \sigma_\alpha^2, \phi) + \mathbb{1}_{\{a_{jt}=0\}} \gamma_{jt}^{ex} - \mathbb{1}_{\{a_{jt}>0\}} [C(\Delta_N; \gamma^c) - \nu_{jt}] \right\} - \sum_{j \in J_t^{pe}} \mathbb{1}_{\{a_{jt}^{pe}=1\}} \gamma_{jt}^{en}, \quad (19)$$

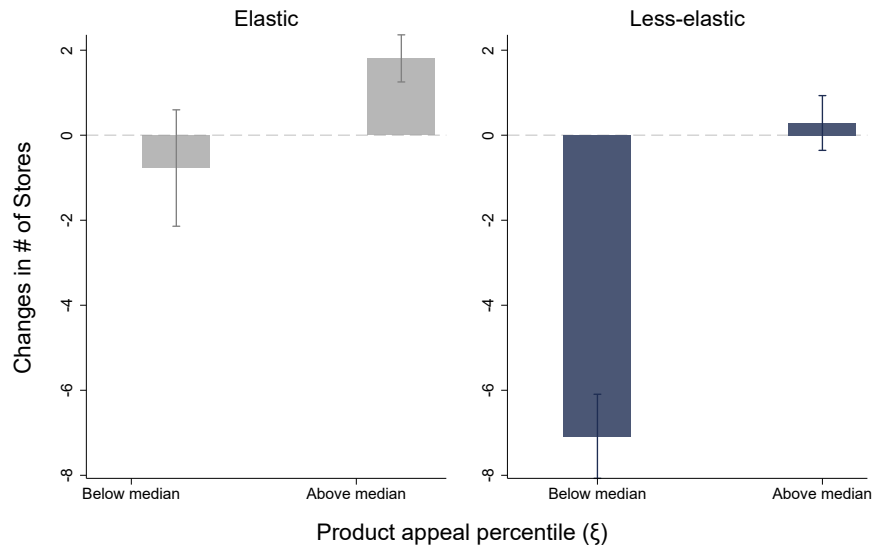
where PS_t represents the expected profit of all producers in the market, net of entry, exit, and adjustment costs. The detailed formulation is provided below.

Based on equations (18) and (19), I compute social welfare, $\mathbf{SW}$, as the sum of consumer

Figure 9: Mean Exit and Market Coverage Adjustment



(A) Exit



(B) Market Coverage Adjustment

Note: Each bar represents the mean exit rate (Panel A) and the average change in the number of stores carrying each product (Panel B) during the Great Recession. Products are divided into below- and above-median groups based on estimated product appeal (ξ) and into elastic versus less-elastic groups according to the recovered product-level price elasticity. The recession period covers 2007Q4–2009Q2. Error bars indicate 90% confidence intervals.

and producer surplus over the entire sample period, pooling across all industries in the sample and applying a social discount factor β_s . The social discount factor is set higher than the private discount factor β to reflect the social planner's longer-term perspective. Specifically, I use $\beta_s = 0.999$ and $\beta = 0.99$. The social welfare is then expressed as

$$\mathbf{SW} = \mathbb{E} \left[\sum_{t \geq 1}^{\bar{T}} \beta_s^{t-1} (CS_t + PS_t) \right]. \quad (20)$$

I conduct an exercise comparing welfare outcomes under two scenarios: the baseline economy with aggregate demand shocks and a counterfactual economy without shocks. For each scenario, I simulate 500 independent paths of industry evolution and compute social welfare, consumer surplus, and producer surplus as described above. Table 4 reports the average welfare across these simulations. The welfare cost of the business cycle is measured as the difference between the two scenarios, and the corresponding standard errors are reported in parentheses. The results indicate that aggregate fluctuations generate a substantial welfare loss of about \$37 billion, which corresponds to approximately 1.5% of pre-recession personal consumption expenditure (PCE) on food. Given that the NielsenIQ data represent only a subset of the Consumer Price Index (CPI) basket and that the current sample includes roughly 15 industries, this estimated welfare cost is considerably large.

The welfare loss arises primarily through reduced consumer surplus, which declines by about \$41 billion, while producer surplus remains relatively stable. Given that consumer surplus accounts for roughly 10% of total social welfare, this represents a sizable impact. The underlying mechanism is that recessions lead to fewer product entries and more exits, resulting in a market composed of fewer but higher-appeal products offered at higher prices. The decline in consumer surplus indicates that the gains from access to higher-appeal products do not fully offset the losses from higher prices and reduced product variety.

6.2 Welfare Costs across Industries

Next, I turn to the exercise at the disaggregated industry level. Figure 10 illustrates the welfare costs across industries with different demand elasticities. Panel A presents the change in consumer

Table 4: Welfare Costs of the Business Cycle

	Social Welfare	Consumer Surplus	Producer Surplus
With shock	2,873.27	260.42	2,612.85
Without shock	2,910.78	301.84	2,608.94
Welfare costs (difference)	-37.51 (0.49)	-41.42 (0.07)	3.90 (0.48)

Notes: The table reports welfare measures under two scenarios: the baseline economy with aggregate demand shocks and the counterfactual economy without shocks. The last row shows the difference (with shock minus without shock). All values are in billions of dollars. Standard errors are reported in parentheses.

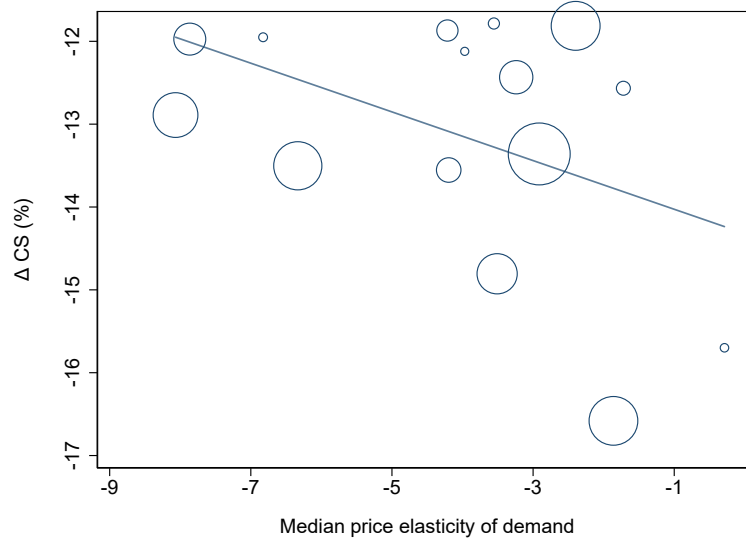
surplus (ΔCS), while Panel B shows the corresponding change in producer surplus (ΔPS). The x-axis in both panels represents the median price elasticity of demand for each industry, derived from the estimated model. The results show that industries with less elastic demand experience larger declines in consumer surplus during recessions. This pattern arises because substitution possibilities are limited in these industries, so consumers continue purchasing similar products even as prices rise, leading to greater welfare losses. For producer surplus, industries with less elastic demand tend to experience slight increases following shocks. This pattern is mainly driven by higher exit rates in those industries, as producers obtain larger exit values and face lower effective entry costs.

6.3 Counterfactual Exercise: No Consumer Heterogeneity

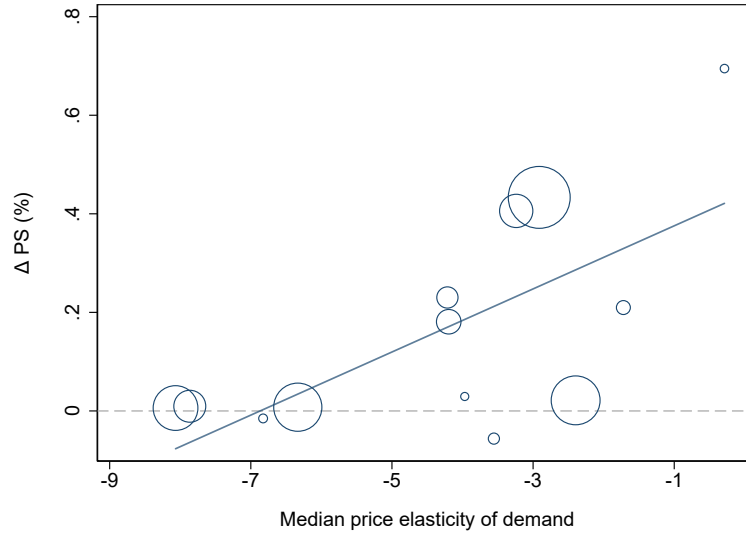
Finally, I examine the role of demand-side heterogeneity in shaping welfare outcomes through product market dynamics. In the baseline model, consumers differ in their price sensitivity within each industry. To assess the importance of this heterogeneity, I conduct a counterfactual exercise setting the variance of price sensitivity to zero ($\sigma_\alpha^2 = 0$), which collapses the random-coefficients logit model into a standard logit demand system, closely resembling the representative-consumer framework in a CES demand model (Anderson, De Palma, and Thisse, 1992). I still allow for differences across industries in the mean price elasticity (α), consistent with a nested CES structure that captures across-industry heterogeneity. This experiment isolates how ignoring within-industry consumer heterogeneity affects the magnitude of the welfare costs of the business cycle.

The results in Table 5 show that the welfare costs of the business cycle are understated by

Figure 10: Changes in Welfare across Industries



(A) Consumer Surplus



(B) Producer Surplus

Notes: Each bubble represents an industry, with bubble size proportional to its revenue. The x-axis shows the median price elasticity of demand derived from the estimated model. ΔCS and ΔPS denote percentage changes in consumer surplus and producer surplus, respectively, between the baseline (with shock) and the counterfactual (without shock) economies.

Table 5: Welfare Costs under No Consumer Heterogeneity

	ΔSW	ΔCS	ΔPS
Baseline	-37.51 (0.49)	-41.42 (0.07)	3.90 (0.48)
No consumer heterogeneity	-35.59 (0.45)	-39.23 (0.07)	3.64 (0.44)
Compared to baseline (%)	-5.12	-5.27	-6.77

Notes: The table reports welfare costs for each scenario. The last row shows the percentage difference of the counterfactual relative to the baseline. All values are in billions of dollars. Standard errors are reported in parentheses.

roughly 5% when consumer heterogeneity is ignored, equivalent to about 0.2% of pre-recession personal consumption expenditure (PCE) on food. The attenuation arises because a representative consumer model overestimates substitutability across products, thereby dampening the welfare impact of product entry and exit. In contrast, when consumers differ in their preferences, changes in product variety and composition have stronger welfare implications, as some consumers are more sensitive to changes in the set of available products through entry and exit. These findings underscore the importance of accounting for heterogeneous consumer responses when evaluating welfare dynamics in differentiated-product markets.

7 Conclusion

This paper investigates how business cycles reshape product market dynamics, focusing on the interaction between consumer demand and producer entry and exit. I develop a dynamic model with heterogeneous producers and consumers and estimate it using detailed micro-level scanner data from the U.S. consumer goods sector. The estimated model provides new evidence on how aggregate demand shocks affect product entry and exit decisions, highlighting the critical role of demand heterogeneity in shaping these dynamics and their welfare implications.

The analysis shows that producers systematically reduce product entry and increase exit during recessions, with the magnitude of these adjustments varying across markets. These differences arise from variation in the price elasticity of demand and product characteristics. For entry, markets with less elastic demand, characterized by higher sunk entry costs, experience particularly large declines

in product introduction. For exit behavior, products with lower consumer appeal and facing less elastic demand are more likely to exit and reduce their market coverage during recessions. This finding is important because, in a standard monopolistic competition framework, the price elasticity of demand is often viewed as the inverse of the markup, suggesting greater market power in less elastic markets. However, this paper shows that such an interpretation can be misleading: less elastic demand is associated with higher markups only for high-appeal products that remain in the market, whereas low-appeal products facing limited substitution possibilities are more likely to exit despite charging lower prices.

These dynamics imply significant welfare losses for consumers, primarily through higher prices and reduced product variety. The welfare cost of the business cycle is substantial, and this decline in welfare is largely driven by the reduction in consumer surplus. Moreover, the magnitude of welfare losses varies systematically across industries, with those facing less elastic demand experiencing larger declines. Finally, ignoring consumer heterogeneity leads to an underestimation of welfare costs by about 5%, highlighting the importance of accounting for diverse consumer preferences when evaluating welfare dynamics in differentiated-product markets.

The heterogeneity in product market responses underscores the importance of accounting for rich demand structures when analyzing business cycle effects. Future research could extend this framework to examine how consumer demand itself evolves over the business cycle, particularly through trading-down behavior, or to incorporate non-homothetic preferences into the model. Overall, this paper highlights the importance of studying product market dynamics as a critical channel through which business cycles affect welfare.

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Online Appendices

Figure 1: Number of UPCs within Brand

